

***Voice Äpp*: A smartphone application for crowdsourcing Swiss German dialect data**

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1 Introduction

The spatial variability found in dialects is an essential indexical property that is highly salient to listeners in everyday language situations: at social events, for example, one often hears conversations of the type “I have trouble localizing your dialect – where do you come from?”. Although listeners are typically unaware of the underlying linguistic mechanisms involved, they are actively engaging in perceptual dialectology (cf. Preston 1989, Clopper & Pisoni 2004) and they seem keenly aware of dialectal variation. It is interesting then that different language speaking groups seem to recognize dialects of their language with different degrees of accuracy. Leemann & Siebenhaar (2008) and Guntern (2011) show that naïve Swiss German listeners can accurately recognize a speaker’s dialect with a recognition rate of 86% and 74% respectively. However, Clopper & Pisoni (2005) report identification rates of only 30–50% for American and British English dialects; Kehrein, Lameli & Purschke (2011) report similar recognition rates for German dialects. Recent studies show that dialect recognition is possible via the mobile application *Dialäkt Äpp* (Leemann & Kolly, 2013; Kolly & Leemann, in review).

This contribution describes work in progress: *Voice Äpp*, currently in development at the University of Zurich, is a follow-up project on *Dialäkt Äpp*. The main purpose of both smartphone apps is to identify users’ dialects on the basis of the dialectal variants of 16 words. *Dialäkt Äpp* users provide their pronunciation through tapping on the corresponding variant on the smartphone screen. However, the new *Voice Äpp* asks users to pronounce the word and uses automatic speech recognition (ASR) to identify users’ pronunciation variants. The ASR training for *Voice Äpp* is partly based on acoustic data crowdsourced through *Dialäkt Äpp*. *Voice Äpp* further aims at illustrating the individuality in users’ voices by providing a multidimensional profile of their voice. The launch of *Voice Äpp* is planned in December 2014.

Several research teams are interested in creating similar applications for other languages, using the frameworks put forth by *Dialäkt Äpp* and *Voice Äpp*: Mobile applications that recognize regional varieties of the entire German-speaking area, of American English, of British English, and of Italian, are currently under development.

2 Crowdsourcing data with *Dialäkt Äpp*

In 2013 we launched the iOS application *Dialäkt Äpp*, which capitalizes on the Swiss public interest in dialectology (Leemann & Kolly, 2013). We provided a functionality that, on the one hand, allows users to localize their own Swiss German dialect by indicating their pronunciation of 16 words (see Figure 1). Given the task to predict Swiss German dialects, a model was built by phoneticians who devised a set of maximally predictive words (i.e. maps from the Linguistic Atlas of German-speaking Switzerland: *Sprachatlas der Deutschen Schweiz* (SDS, 1962–2003)) that capture dialectal differences between localities. On the other hand, users can record their own dialect and listen to recordings of other users, thus discover the Swiss dialectal landscape. Figure 1 shows three screens of the application: the choice of dialectal variants for the word *Donnerstag* ‘Thursday’, the identified localities as a

list and on a map (*Bern* being the best hit in this example) and the distribution of users' recordings covering German-speaking Switzerland.



Figure 1: Screens of *Dialäkt Äpp*: (1) choice of dialectal variants with buttons; (2) result provided as a choice of five best hits and their corresponding positions on a map; (3) users' recordings (one pin per locality)

Dialäkt Äpp was launched on March 22, 2013, and has been downloaded over 58'000 times (as of February 28, 2013). The data recorded by this application contains (a) (written) choices of pronunciation for 16 words by each user who localized his/her dialect and (b) audio data for the same 16 words by each user who chose to record his/her voice. For (a), the corpus contains data from over 42'000 subjects (58% males, 42% females). Most users are from the cantons (and capitals) of Zurich, Bern, Basel, Luzern, Aargau, and St. Gallen. 64% of the users' pronunciation variants still correspond to the local variant recorded by the SDS (1962–2003) in the 1940's and a large number of users report that the localization of their dialect by the application is very close to their dialectal origin. For (b), the corpus counts 38'477 recorded variants stemming from a total number of 2'633 iOS devices (which corresponds roughly to the number of speakers; 54% males, 46% females). The geographical distribution of users corresponds to that of the data presented in (a).

The data elicited by *Dialäkt Äpp* has great potential for dialectological as well as forensic phonetic research. It can be used to create new dialect maps and compare them to the maps published in the SDS (1962–2003), thus to track sound change in progress. A number of maps have already been created (for the words *Apfelüberrest* 'apple core', *Bett* 'bed', *schneien* 'to snow', *Tanne* 'pine tree', and *tief* 'low'). Preliminary analyses show that phonetic isoglosses, as illustrated in maps like *Bett* (quality of /e/) and *Tanne* (quantity of /n/) are congruent with data from the SDS (1962–2003) (Kolly & Leemann, in review). The data can also be used to compare dialects at the acoustic phonetic level: For example, preliminary results show differences in speaking rate between the Bern dialect and the Zurich dialect (Leemann, Kolly, & Dellwo, accepted). Furthermore, this corpus can be used to create population statistics for a variety of phonetic parameters, which is desirable for forensic phonetic voice comparison (cf. Nolan et al., 2009).

3 Development of *Voice Äpp*

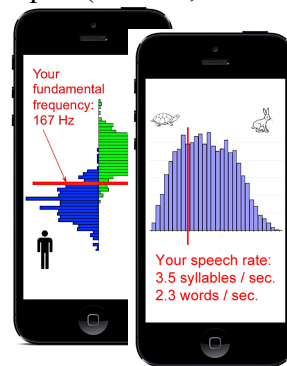
Voice Äpp has two major aims:

- To use ASR techniques to localize users' dialects
- To provide users with a multidimensional profile of their voice

3.1 ASR-based dialect localization

The novelty of this new project is to use ASR techniques instead of multiple choice buttons. Some difficulties can be expected as the ASR approach is not error-free, especially through a mobile application: recording conditions may vary a lot due to the distance from the microphone, noisy environments etc. However, the high-resolution microphones of smartphones, iPhones in particular, should facilitate the ASR task. Furthermore, identifying dialects, where small variation has to be taken into account, is not the initial purpose of ASR systems; the speech recognition domain aims at normalizing such variation and at being rather dialect- or speaker-independent. In addition to this, the number of possible pronunciation variants for each word is important. For example, the word *Bett* ‘bed’ only counts two variants in the SDS (/bet/ and /bet/) whereas *Augen* ‘eyes’ has eleven dialectal pronunciation variants. The latter is highly discriminant – but the ASR task is more difficult. The algorithm will have to be modified since the voice recognition approach is not as reliable as the selection with buttons.

In order to achieve this, an ASR system is trained with two corpora: (a) the *Dialäkt App* corpus described in 2 and (b) the TEVOID corpus (Dellwo, Leemann, & Kolly, 2012). Corpus (a) contains about six hours of speech of speakers, covering a dense net of local dialects speaking Switzerland. Each recording is an from a set of 16 words. Corpus (b) contains two minutes of speech of 16 Zurich German recording is either a spontaneous or a read the second corpus has been segmented by hand, needs data preparation and verification as it was control of linguistic content nor acoustic



over 2'600 in German-isolated word hours and 45 speakers. Each sentence. While the first one collected without environment.

So far, encouraging results are obtained with limited training data. After ASR training with five variables from the *Dialäkt App* corpus, dialect word recognition has reached accuracies of 92% (*Bett* ‘bed’), 90% (*Kind* ‘child’, *Apfelüberrest* ‘apple core’), 85% (*Tanne* ‘fir tree’), 79% (*fragen* ‘to ask’). These accuracies may increase with larger amounts of training data, which is currently being worked on.

3.2 Multidimensional voice profile and infotainment content

The second function of the *Voice App* is a voice profile provided to the user. Based on a sentence recorded in their dialect, users learn about characteristics of their own voice in a playful way. A number of menus allow users to explore different aspects of speech, e.g. pitch, speech rate, articulation, auditory and visual perception.

Pitch: The fundamental frequency (f0) of the users’ sentence is calculated and displayed in a histogram representing the distribution of the f0 of all the previous users.

Speech rate: The speech rate of the users’ sentence is calculated and displayed in comparison to the previous users’ speech rate.

Articulation: Users learn about sounds and their articulation. Upon clicking on an IPA symbol a sagittal cut is shown and the sound is played. In an interactive sagittal cut users move the position of the articulators and hear the corresponding vowel sound.

Auditory perception: Users can listen to what their sentence would sound like to a person with a hearing impairment/a cochlear implant.

Visual perception: Users are shown a video illustrating the *McGurk effect* (MacDonald & MacGurk, 1978) and the *Cocktail Party Effect* (Handel, 1989). Both effects illustrate that visual cues can be crucial for speech perception.

4 Conclusion

Voice Äpp should be as interactive as possible, allowing users to learn about the individual features of their dialect and their voice in a playful way. As shown by *Dialäkt Äpp*, a mobile application such as *Voice Äpp* is interesting for the user as well as for the researcher: by providing appealing content to the user, we gain large amounts of data. This crowdsourced data can be used to create population statistics, for example for analyses of speech prosodic features. In particular, *Voice Äpp* creates real time f0 and speaking time statistics, which represents a novelty for e.g. the field of forensic phonetics.

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